ECE 340: Semiconductor Devices

Lecture 25: P-N Junction Under Bias

Prof. Hyunseok Kim

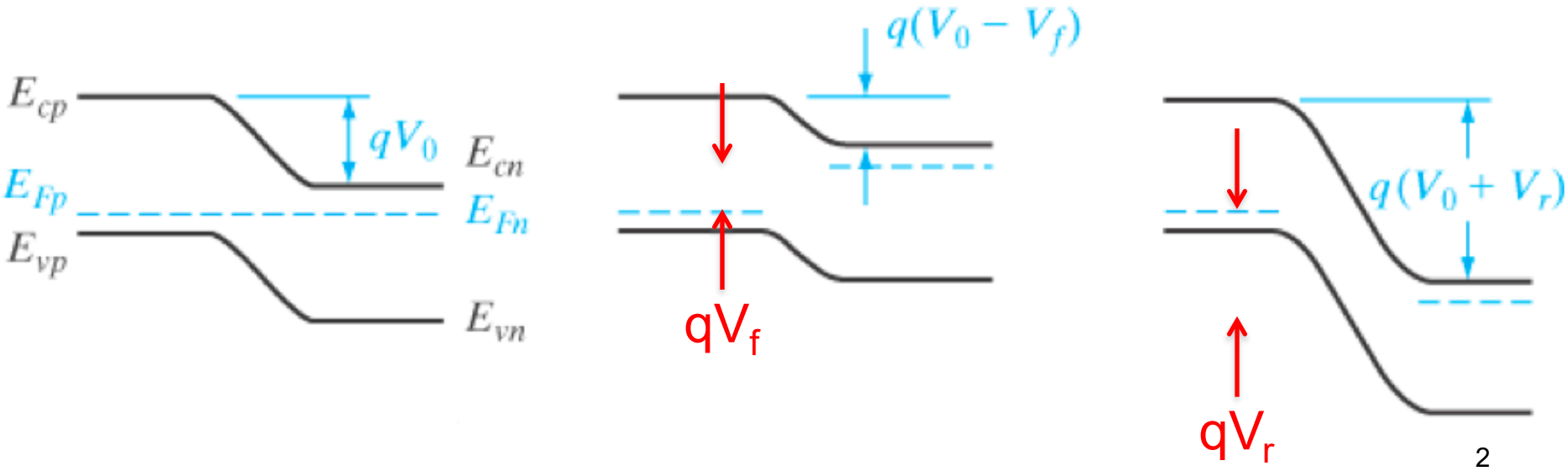
Lecture (Section C): M,W,F / 2:00-2:50pm / 3017 ECEB

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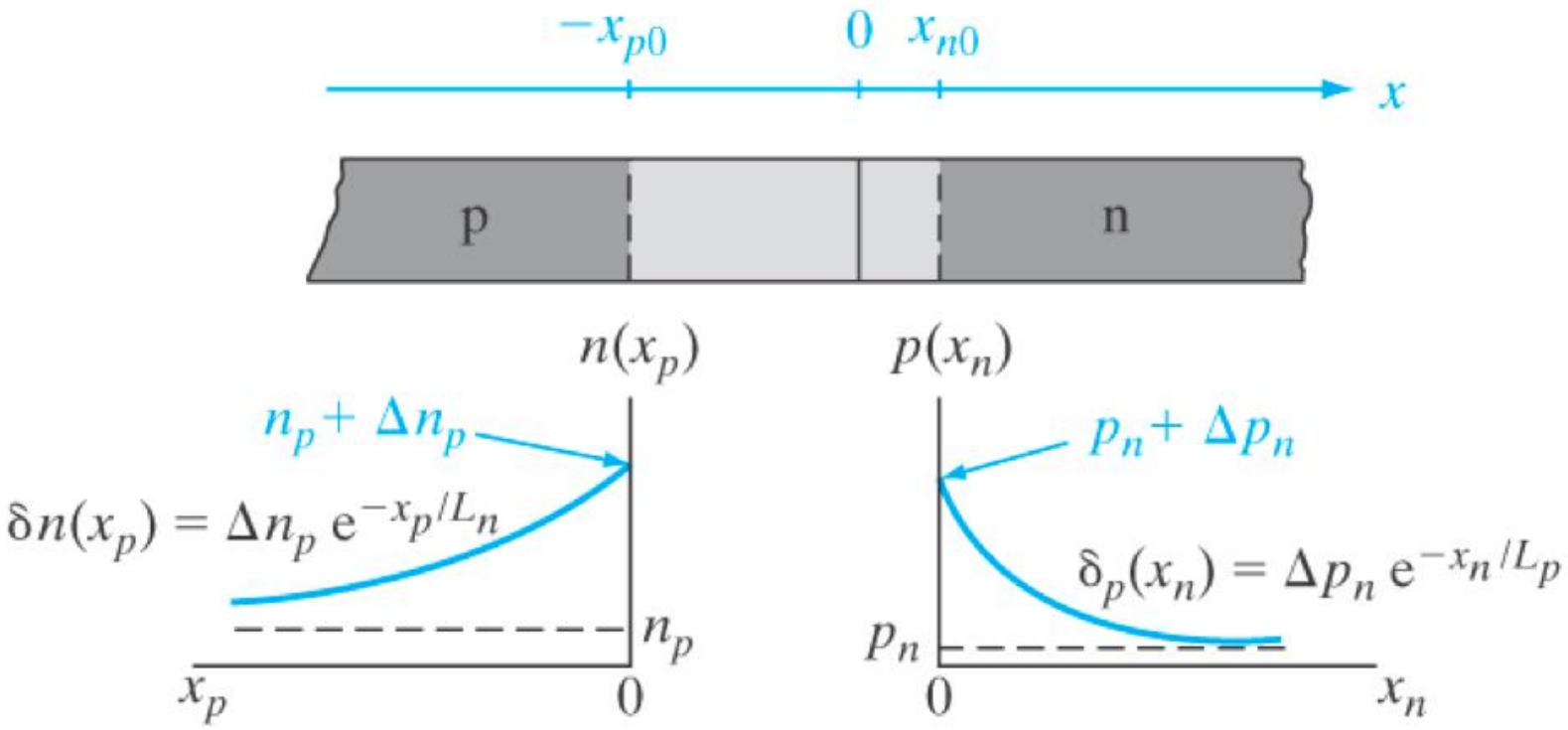
Office Hours: Thursdays 12p-1p, 2256 HMNTL

(Recap) Bias and Fermi Level Separation

- When we apply voltage we are no longer in equilibrium
- The application of a voltage creates a difference in the electrochemical potential across the junction boundary
- The Fermi levels become separated by an energy of "q" times the applied voltage
- The band separation is $q(V_0-V)$



(Recap) The Case We Need to Examine



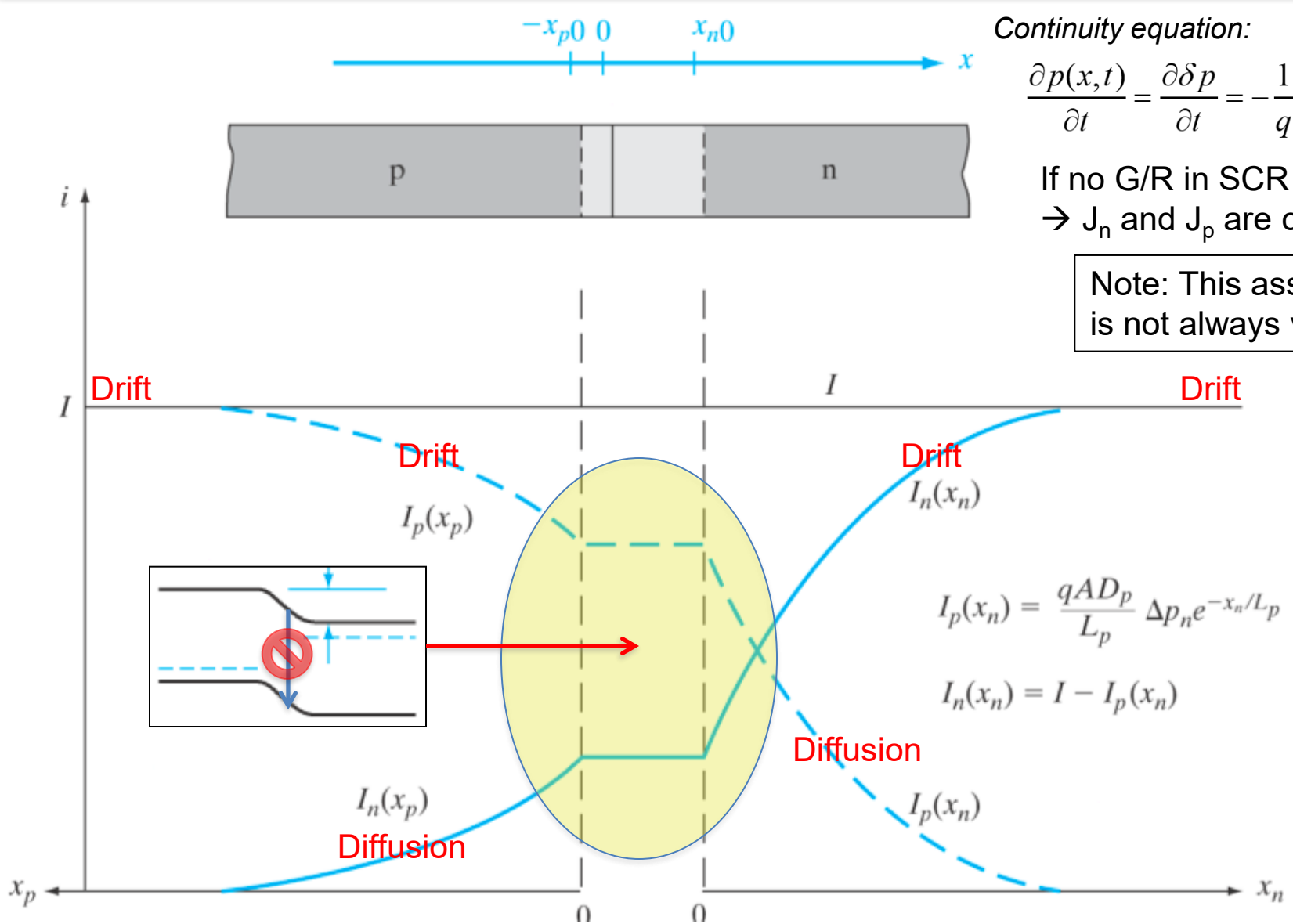
Coordinate Transformation:

$x_n = x - x_{n0}$

$x_p = -x - x_{p0}$

x_{n0} and x_{p0} are the n and p-side depletion widths

(Recap) No Recombination in Depletion Region



Continuity equation:

$$\frac{\partial p(x,t)}{\partial t} = \frac{\partial \delta p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} - \frac{\delta p}{\tau_p} + G$$

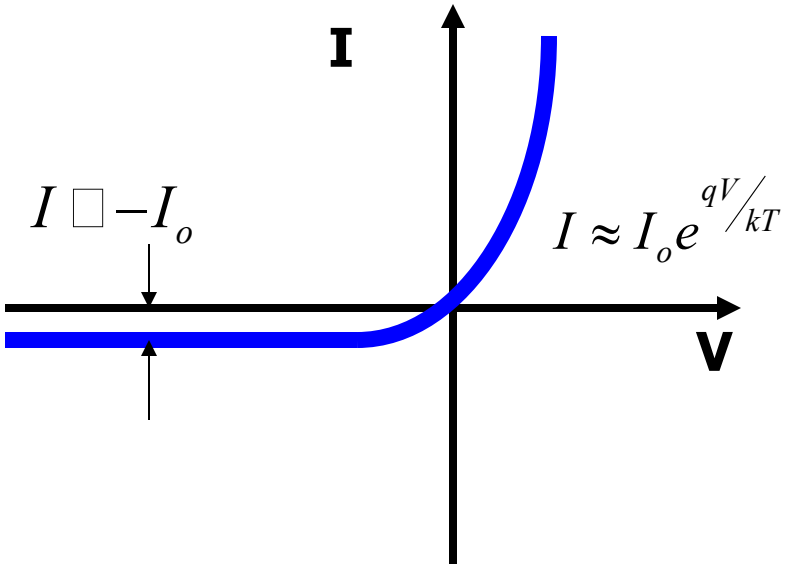
If no G/R in SCR region
→ J_n and J_p are constant!

Note: This assumption is not always valid.

$$I_p(x_n) = \frac{qAD_p}{L_p}\Delta p_n e^{-x_n/L_p}$$

$$I_n(x_n) = I - I_p(x_n)$$

(Recap) Diode Equation: Various Conditions



$$I = I_o(e^{qV/kT} - 1)$$

$$I \approx I_o e^{qV/kT} \text{ for } V = V_f$$

$$I \approx -I_o \text{ for } V = -V_r$$

$$I = 0 \text{ for } V = 0$$

$$I = qA\left(\frac{D_p P_{on}}{L_p} + \frac{D_n n_{op}}{L_n}\right)(e^{qV/kT} - 1) = I_o(e^{qV/kT} - 1)$$

$(a)\, p - n \text{ Junction}$

$$I = qA\left(\frac{D_p P_{on}}{L_p} + \frac{D_n n_{op}}{L_n}\right)(e^{qV/kT} - 1)$$

$(b)\, p^+ - n \text{ Junction}$

$$p_{op} >> n_{on} \Rightarrow n_{op} << p_{on}$$

$$I = qA\left[\frac{D_p p_{on}}{L_p}\right](e^{qV/kT} - 1)$$

$(c)\, p - n^+ \text{ Junction}$

$$p_{op} << n_{on} \Rightarrow n_{op} >> p_{on}$$

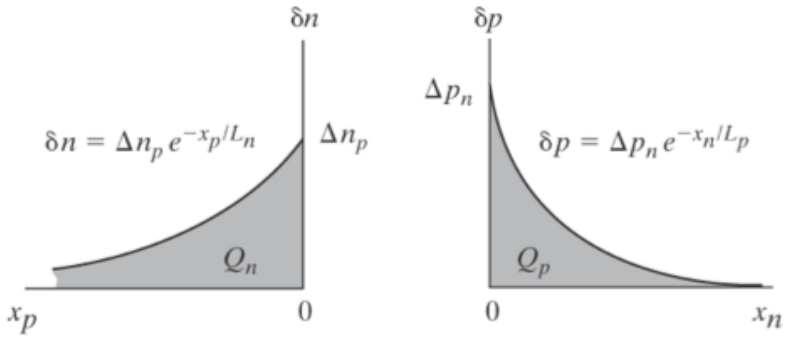
$$I = qA\left[\frac{D_n n_{op}}{L_n}\right](e^{qV/kT} - 1)$$

Charge Control Model

Another Derivation of the Ideal Diode Equation

Charge Control Model

- In steady state, the rate at which we replenish charge across the x_n=0 or x_p=0 must equal the amount of minority carrier charge lost through recombination in an interval of time
- The rate at which charge is lost is equal to the total amount of charge divided by the charge lifetime



$$Q_p = qA\int_0^\infty \delta p(x_n)\,dx_n = qA\Delta p_n \int_0^\infty e^{-x_n/L_p}\,dx_n$$
$$= qA\Delta p_n L_p$$

$$Q_n = qA\int_0^\infty \delta n(x_p)\,dx_p = qA\Delta n_p \int_0^\infty e^{-x_p/L_n}\,dx_p$$
$$= qA\Delta n_p L_n$$

Therefore,

$$I_n(x_p{=}0) = \frac{Q_n}{\tau_n} = \frac{-qAL_n}{\tau_n}\,\Delta n_p$$

$$I_p(x_n{=}0) = \frac{Q_p}{\tau_p} = \frac{qAL_p}{\tau_p}\,\Delta p_n$$

Charge Control Model

Original method (Diffusion current)

$$I_n(x_p=0) = qAD_n \left.\frac{d\delta n}{dx_p}\right|_{x_p=0}$$

$$= -qA \frac{D_n}{L_n} \Delta n_p$$

$$I_p(x_n=0) = -qAD_p \left.\frac{d\delta p}{dx_n}\right|_{x_n=0}$$

$$= qA \frac{D_p}{L_p} \Delta p_n$$

Charge control model

$$I_n(x_p=0) = \frac{Q_n}{\tau_n} = \frac{-qAL_n}{\tau_n} \Delta n_p$$

$$I_p(x_n=0) = \frac{Q_p}{\tau_p} = \frac{qAL_p}{\tau_p} \Delta p_n$$

Recall that: $L_n = \sqrt{D_n \tau_n}$ and $L_p = \sqrt{D_p \tau_p}$

so: $\tau_n = \frac{L_n^2}{D_n}$ and $\tau_p = \frac{L_p^2}{D_p}$

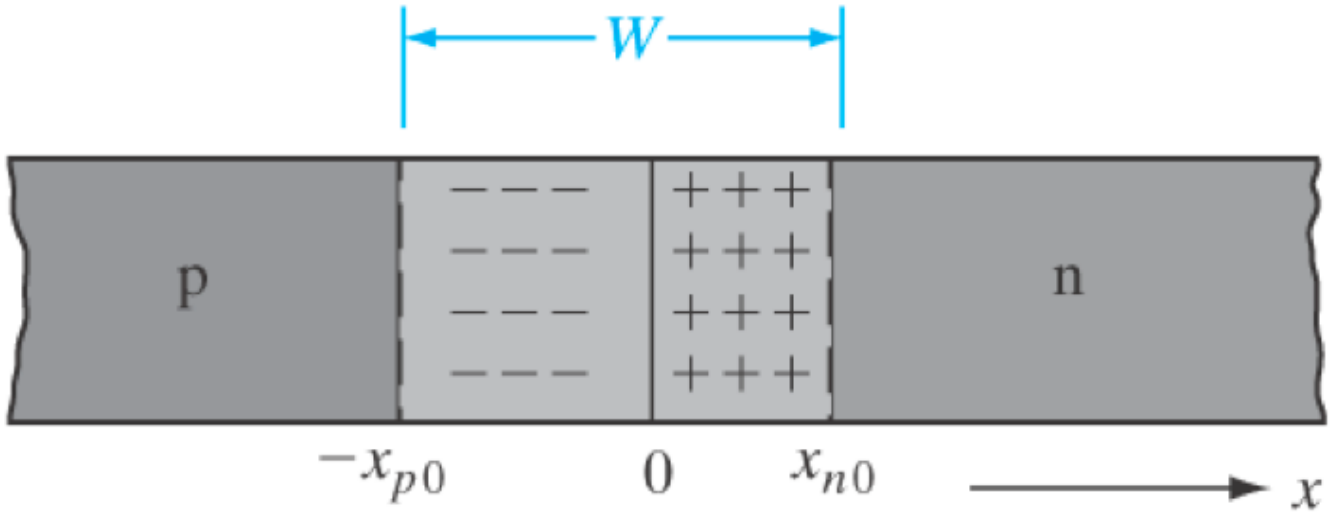
We get the same result:

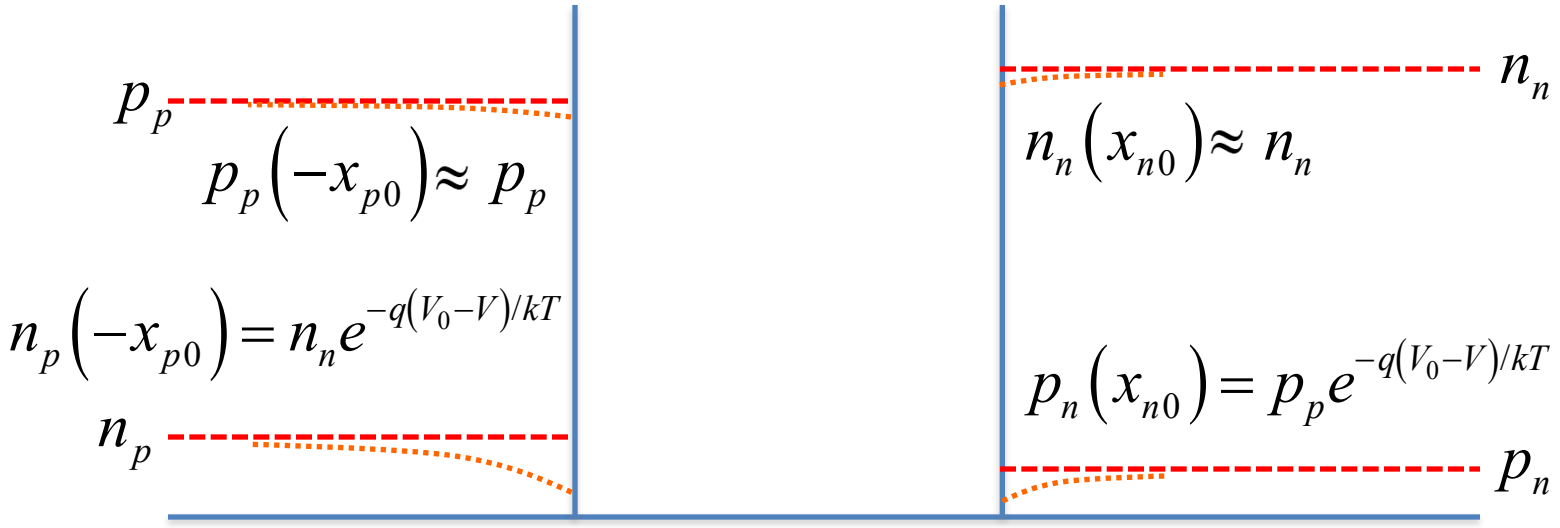
$$I = I_p(x_n=0) - I_n(x_p=0) = qA\left(\frac{D_p}{L_p} \Delta p_n + \frac{D_n}{L_n} \Delta n_p\right)$$

$$= qA\left(\frac{D_p p_n}{L_p} + \frac{D_n n_p}{L_n}\right)(e^{qV/kT} - 1)$$

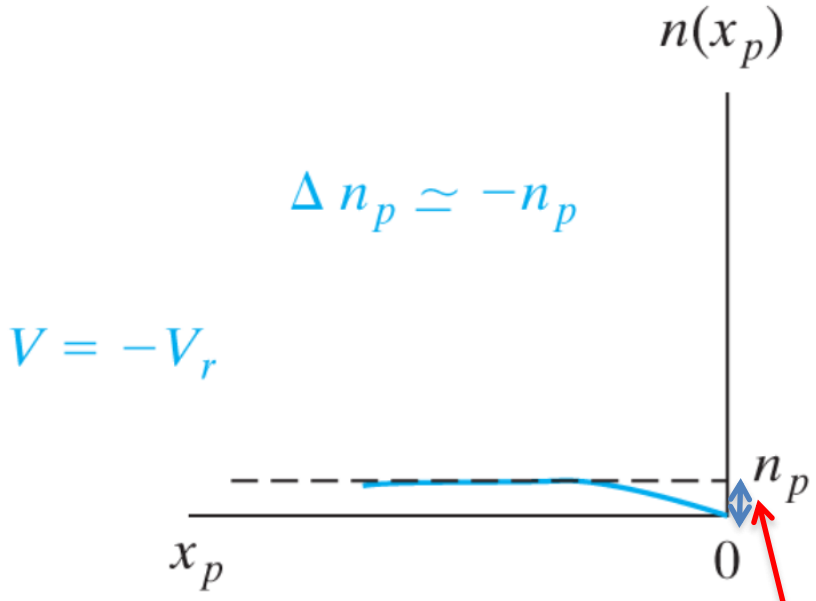
Reverse Bias

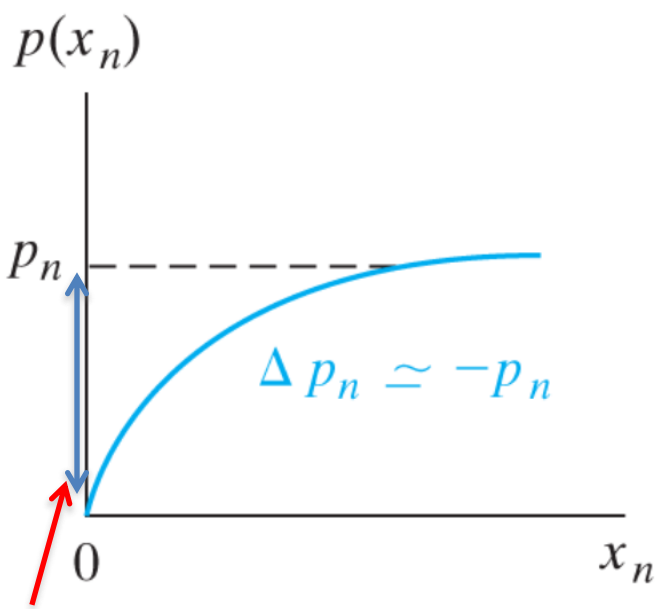
Reverse Bias Charge Distribution





Reverse Bias: Drift Term Dominant





$$n_p\left(-x_{p0}\right) - n_p\left(-\infty\right) = \Delta n$$

$$n_p e^{qV/kT} - n_p =$$

$$n_p\left(e^{qV/kT} - 1\right) =$$

$$-n_p \approx$$

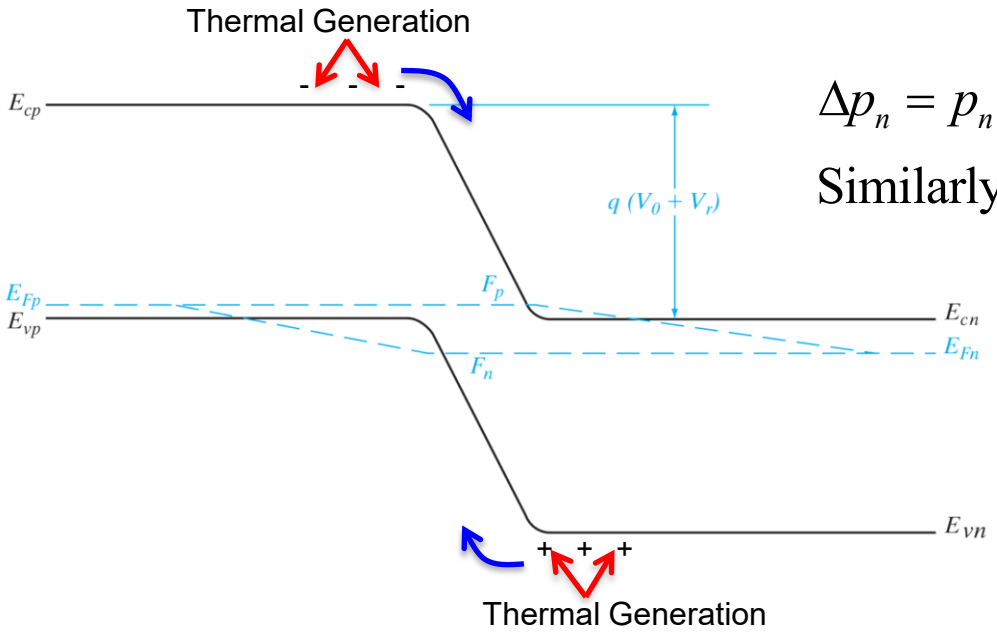
$$\Delta p = p_n\left(x_{n0}\right) - p_n\left(\infty\right)$$

$$= p_n e^{qV/kT} - p_n$$

$$= p_n\left(e^{qV/kT} - 1\right)$$

$$\approx -p_n$$

Reverse Bias Quasi Fermi Levels



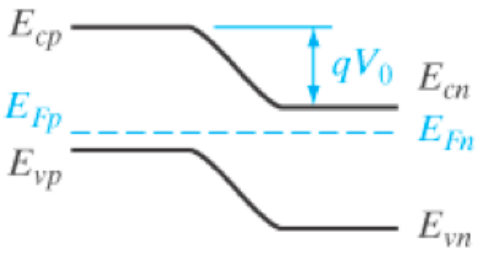
$$\Delta p_n = p_n\left(e^{q(-V_r)/kT} - 1\right) \approx -p_n \quad \text{for} \quad V_r \gg kT/q$$

Similarly: $\Delta n_p \approx -n_p$

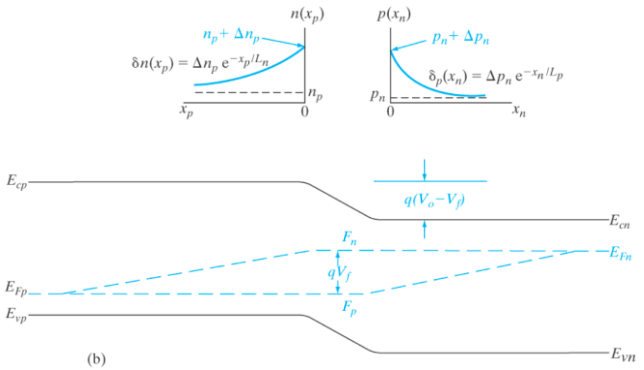
- Minority carrier concentration approaches zero at the edge of the depletion region
- This effect is termed “minority carrier extraction”
- Thermally generated carriers ‘random walk’ to SCR, and be swept away.
- The reverse saturation current is a drift-dominated current, but the minority carriers arrive at the depletion region by diffusion

Junction Currents: Revisited

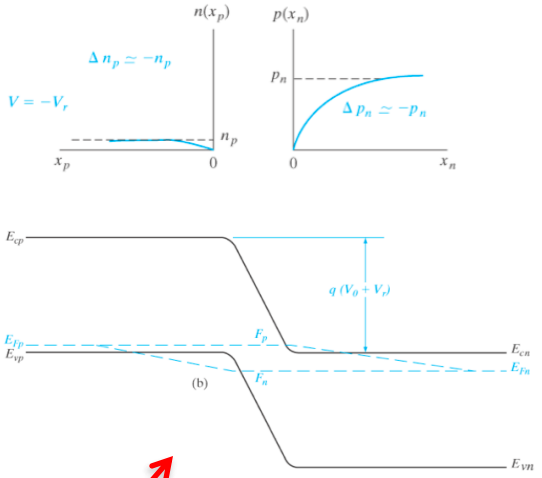
Equilibrium

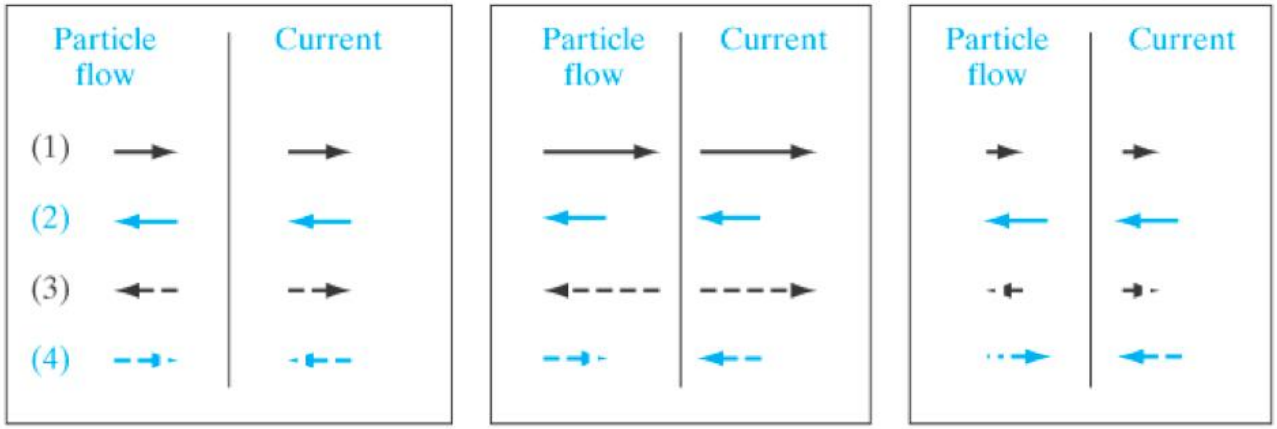


Forward bias



Reverse bias





(1) Hole diffusion
(2) Hole drift
(3) Electron diffusion
(4) Electron drift

Reverse Breakdown

Breakdown Diode

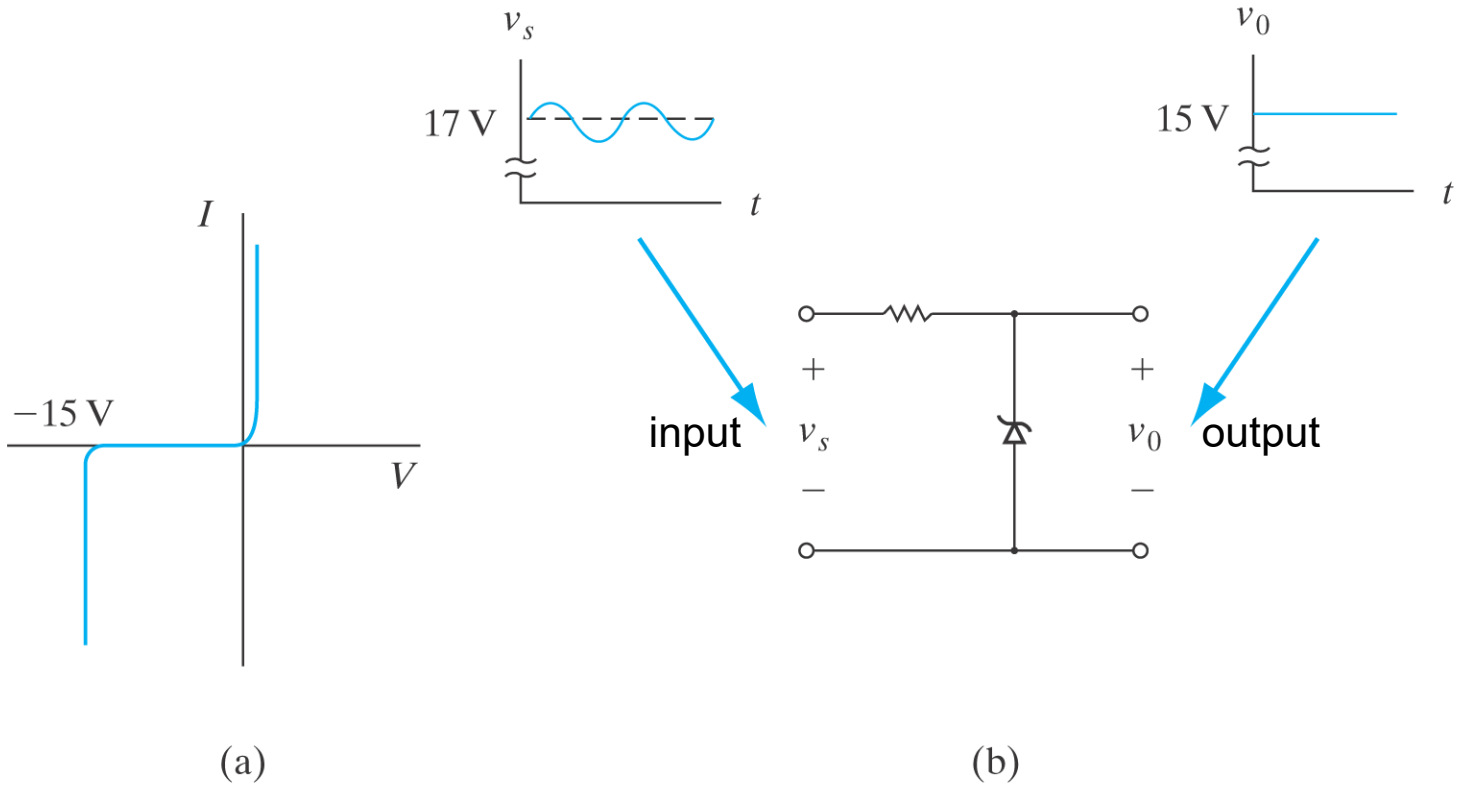


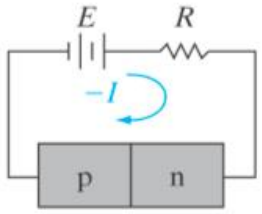
Figure 5.26

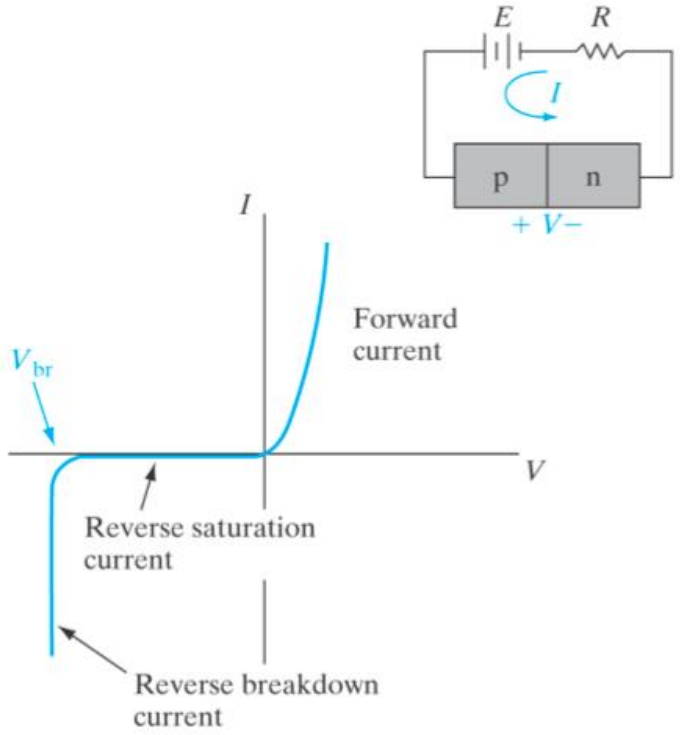
A breakdown diode: (a) *I–V* characteristic; (b) application as a voltage regulator.

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Reverse Breakdown

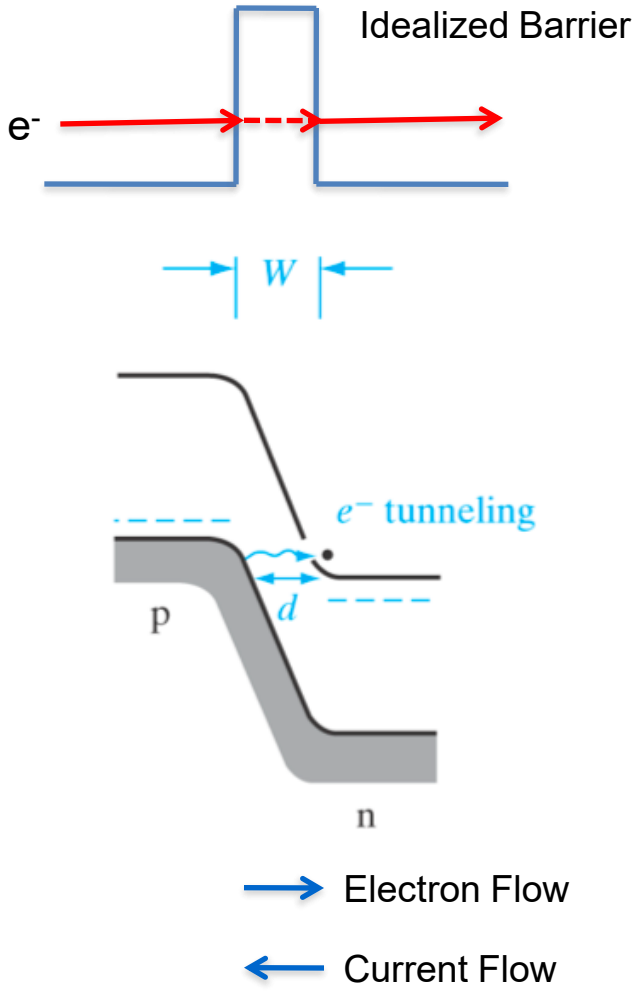
- Reverse “breakdown” is not a destructive process (despite the name) – as long as current is limited externally
- The two types of breakdown discussed here are Zener breakdown and Avalanche breakdown
- **<u>Zener Breakdown</u> occurs at a few volts**
- **<u>Avalanche Breakdown</u> occurs at higher voltages**





Tunneling: A Quantum Effect

- In classical mechanics, a particle can only overcome a potential barrier if it has higher energy than the barrier
- In quantum mechanics, a particle can tunnel through a barrier
- Tunneling probability is enhanced if the barrier is thin, or under certain resonant conditions
- The Zener effect is a tunneling phenomenon



Zener Effect and Zener Breakdown

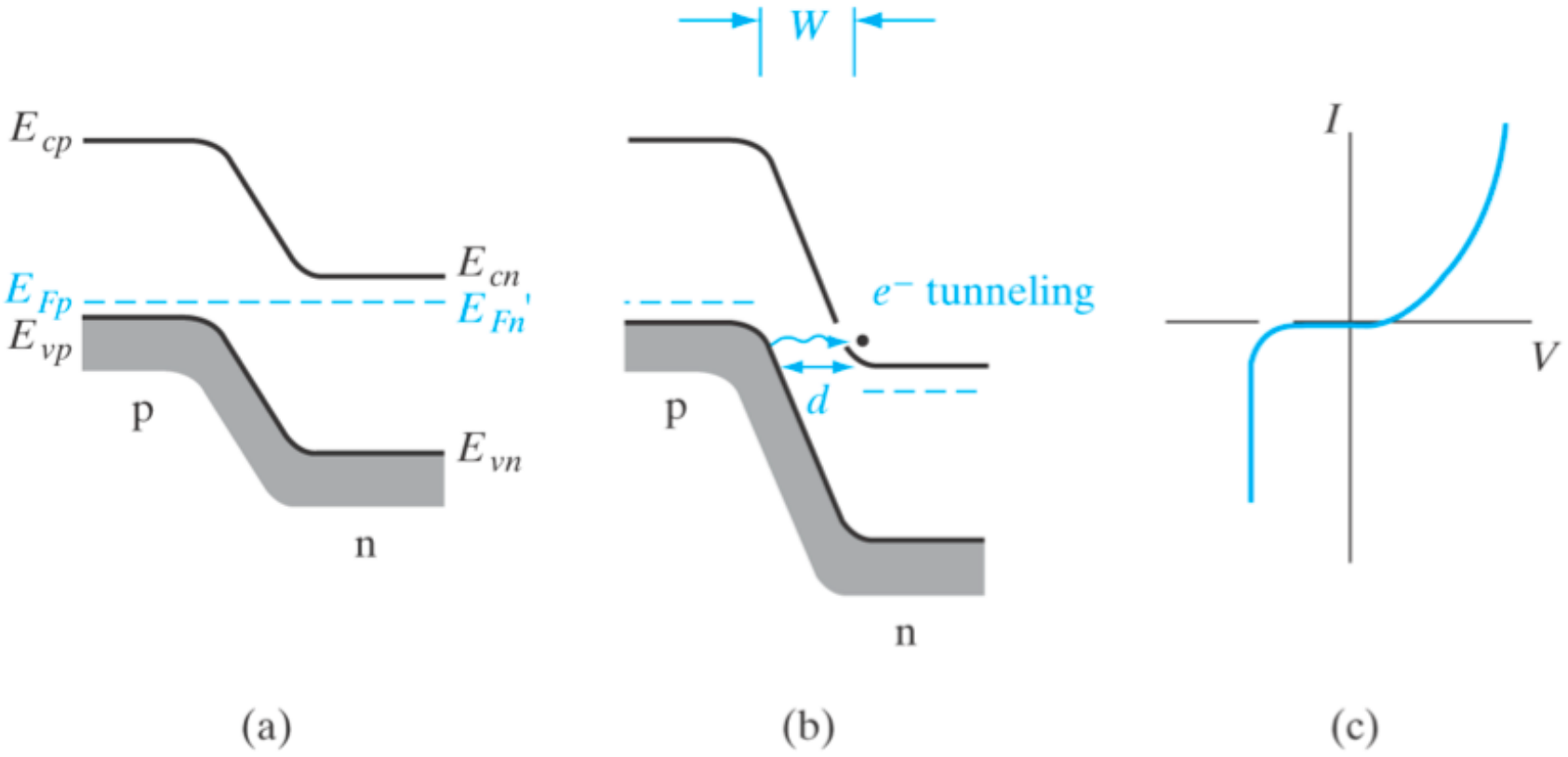


Figure 5.20

The Zener effect: (a) heavily doped junction at equilibrium; (b) reverse bias with electron tunneling from p to n; (c) *I–V* characteristic.

Why is high doping needed?

Zener Effect Comments

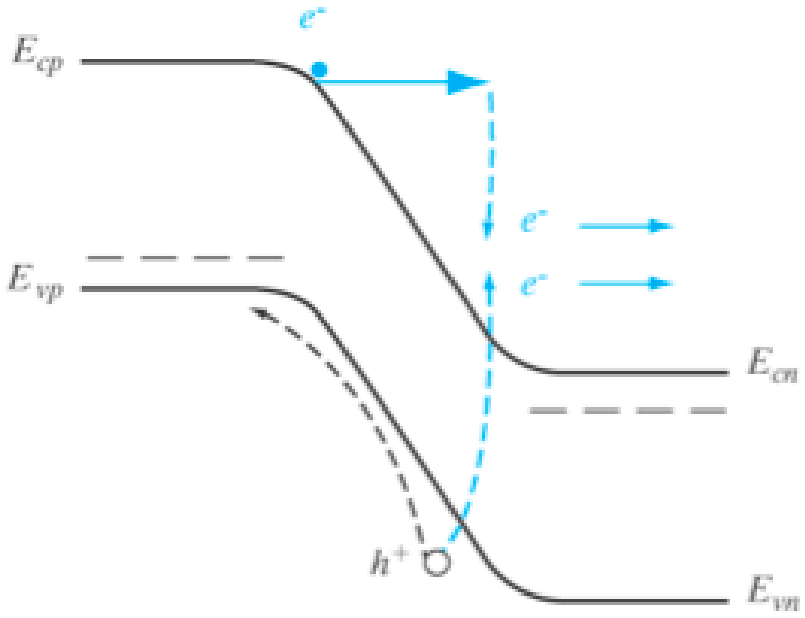
- Tunneling currents can be established when a large number of electrons are separated from a large number of empty states by a narrow barrier of finite height
- The junction must be sharp, and the doping high (small transition region)
- Increasing voltage will produce steeper bands, decreasing the tunneling distance
  - There is some offsetting effect due to a larger depletion width, but at high doping this is small
- This can also be thought of as field ionization of the host atom covalent bonds under high field (~10^6 V/cm)

Reverse Breakdown

Avalanche Multiplication

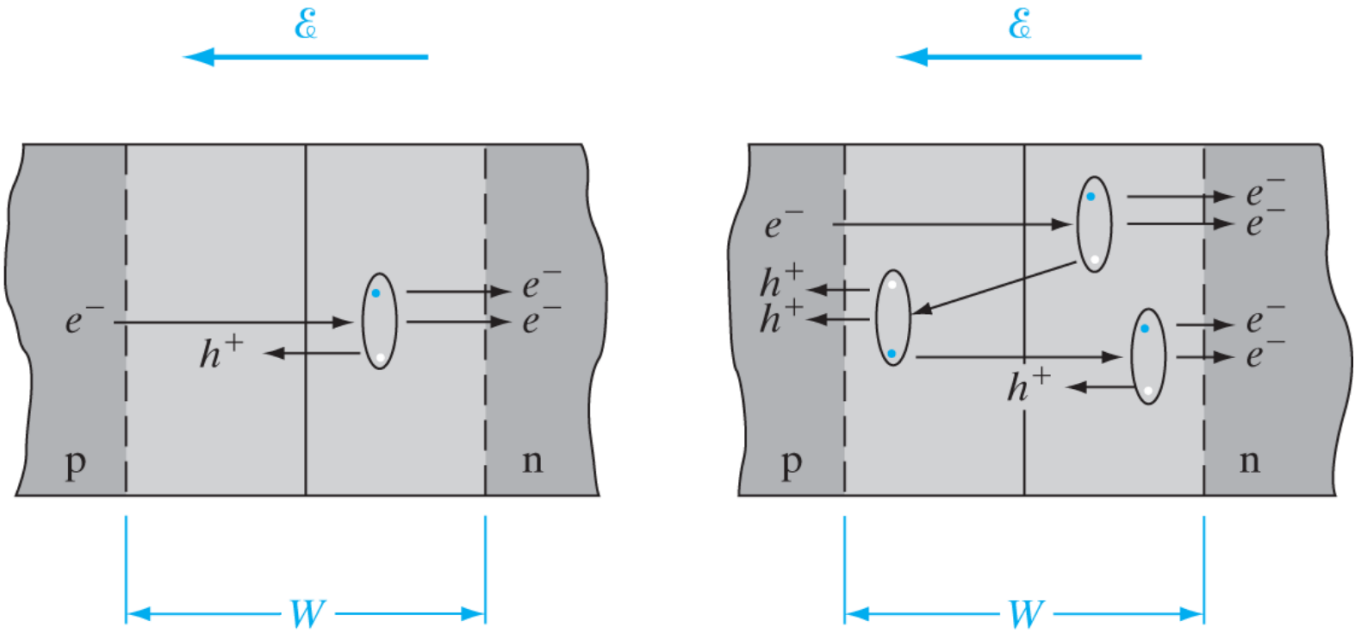
Impact Ionization and Avalanche Breakdown

Impact Ionization



- An electron entering the depletion region is accelerated under high field conditions
- If enough energy is gained, an ionizing collision with the lattice can produce an electron-hole pair ("impact ionization")
- Carrier multiplication occurs ($1\ e^- \rightarrow 2\ e^- + 1\ h^+$)

Impact Ionization and Avalanche Breakdown



- Impact ionization can also occur for the electron or hole created by the first impact ionization event
- This is an avalanche process – a single carrier can produce a large number of electron-hole pairs

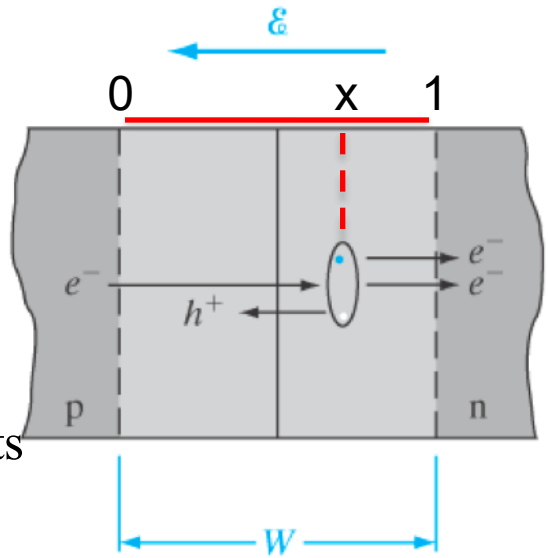
Avalanche Breakdown: Mathematical Treatment

Assume either carrier type has a probability "P" of having an ionizing collision over a distance "W":

1) For n_{in} electrons entering, Pn_{in} electrons will be created and the total number of electrons exiting will be $n_{in} + Pn_{in}$ (1st event)

2) Each created EHP travels a combined distance of "W" wherever created in the depletion region - the electron travels (1-x)W, the hole travels (x)W where "x" is distance normalized to the depletion width W

$P_{h2} = Px$



$P_{e2} = P(1-x)$

P(A OR B) = P(A) + P(B)
when A and B are decoupled events

$P_2 = Px + P(1-x) = P$

Avalanche Breakdown: Mathematical Treatment (2)

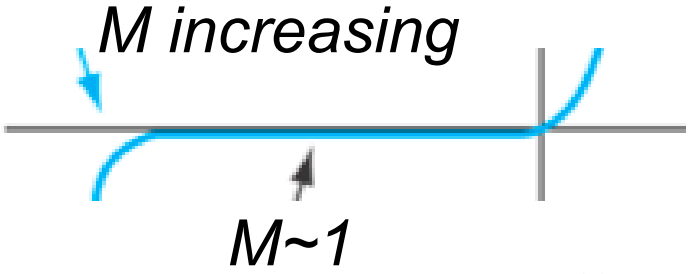
3) The probability of an ionizing collision for secondary carriers is therefore still "P" - for Pn_{in} secondary pairs there will be $P(Pn_{in})$ collisions and $P^2 n_{in}$ tertiary pairs, for $P^2 n_{in}$ tertiary pairs there will $P(P^2 n_{in})$ collisions and $P^3 n_{in}$ quaternary pairs, etc.

Therefore: $n_{out} = n_{in} + Pn_{in} + P^2 n_{in} + P^3 n_{in} + \ldots = n_{in}(1 + P + P^2 + P^3 + \ldots)$

$$M_n = \frac{n_{out}}{n_{in}} = 1 + P + P^2 + P^3 + \ldots = \frac{1}{1 - P}$$

The empircal relationship between M, applied voltage V, and V_{br} is:

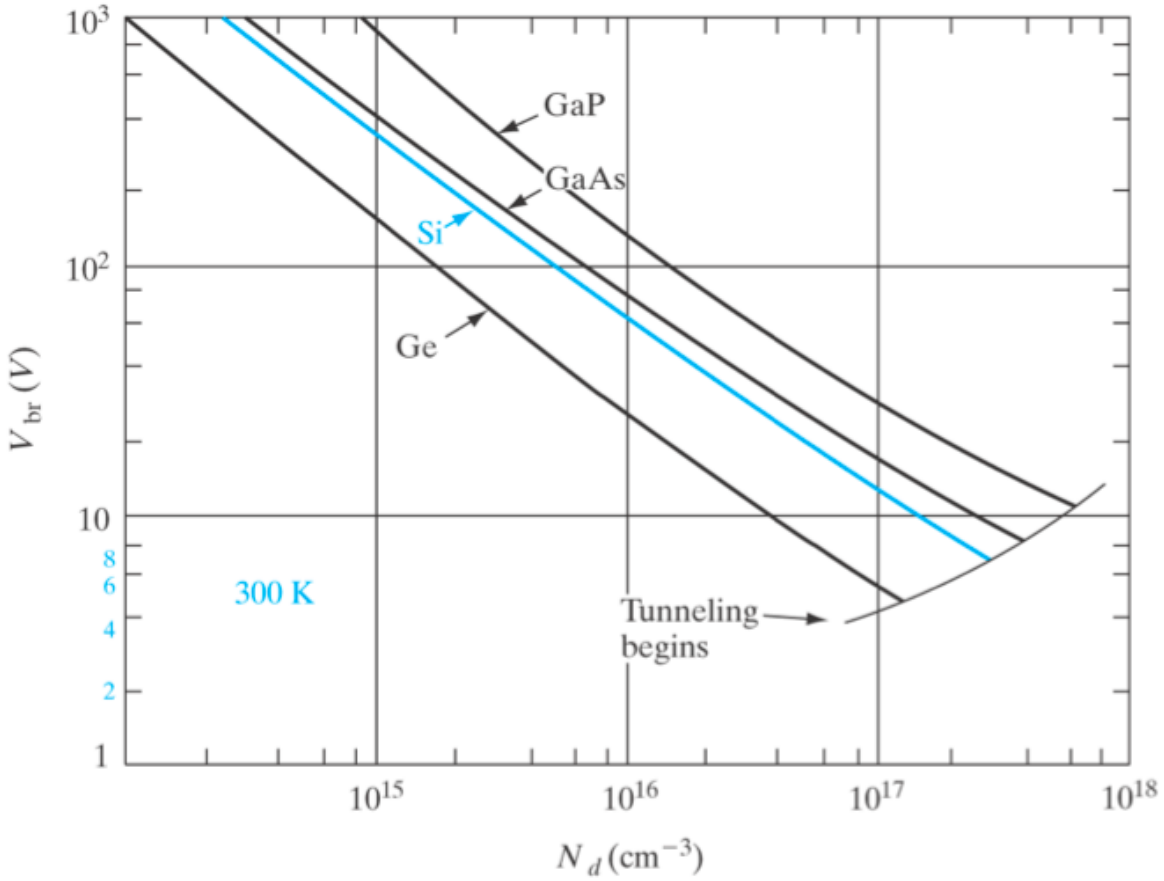
$$M = \frac{1}{1 - (V / V_{br})^n}$$ where n typically varies from 3 to 6



Variation of V_{br} with Material and Doping

Typically:

- V_{br} increases with increasing bandgap
- V_{br} decreases with increasing doping (peak electric field in depletion region is higher)



Capacitance of p-n Junctions

What is Capacitance?

Linear Case

$$C = \frac{Q}{V}$$

- The amount of charge stored for a given potential

Nonlinear Case

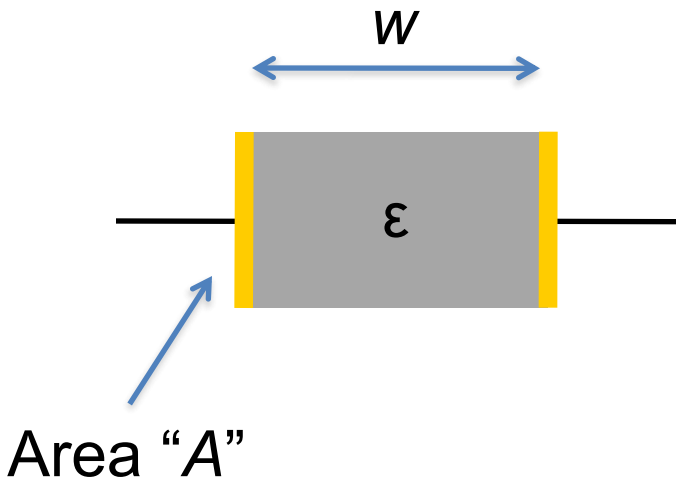
$$C = \frac{dQ}{dV}$$

- The change in charge induced by a change in potential

$$\text{Note: } i = \frac{dQ}{dt} = \frac{dQ}{dV}\frac{dV}{dt} = C\frac{dV}{dt}$$

Reminder: Parallel Plate Capacitance

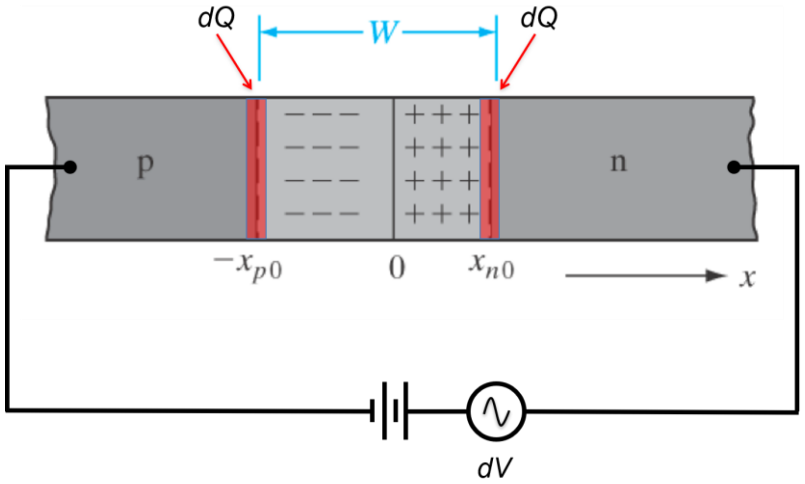
$$C = \frac{\varepsilon A}{w}$$



Two sources of capacitance in p-n junction

Junction (or Depletion) Capacitance:

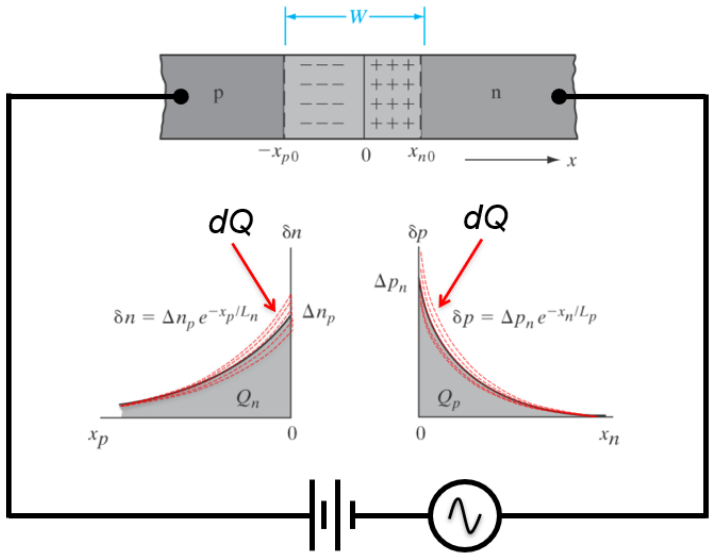
- Due to the dipole in the transition region
- Dominant in reverse bias



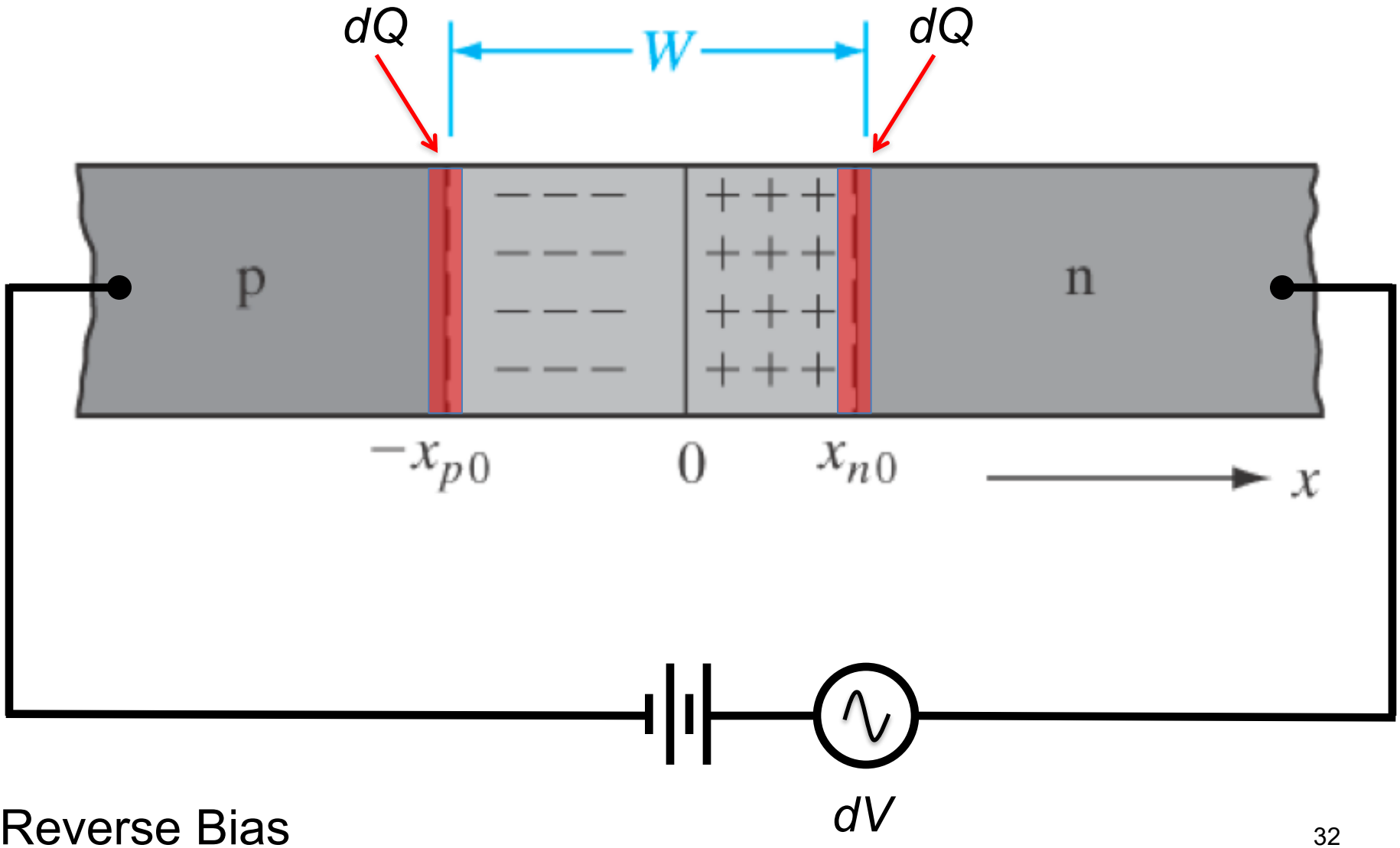
Charge Storage (or Diffusion) Capacitance:

- Due to lagging of voltage as current changes due to charge storage effects
- Dominant in forward bias

More complex than C=Q/V



Depletion/Junction Capacitance



Reverse Bias

Reminder: Calculation of Depletion Width

The negative of the contact potential is the area under the $\mathcal{E}(x)$ versus x triangle:

$$V_o = -\frac{1}{2}\mathcal{E}_o \cdot W = \frac{1}{2}\frac{q}{\varepsilon}N_d x_{no} W = \frac{1}{2}\frac{q}{\varepsilon}N_a x_{po} W$$

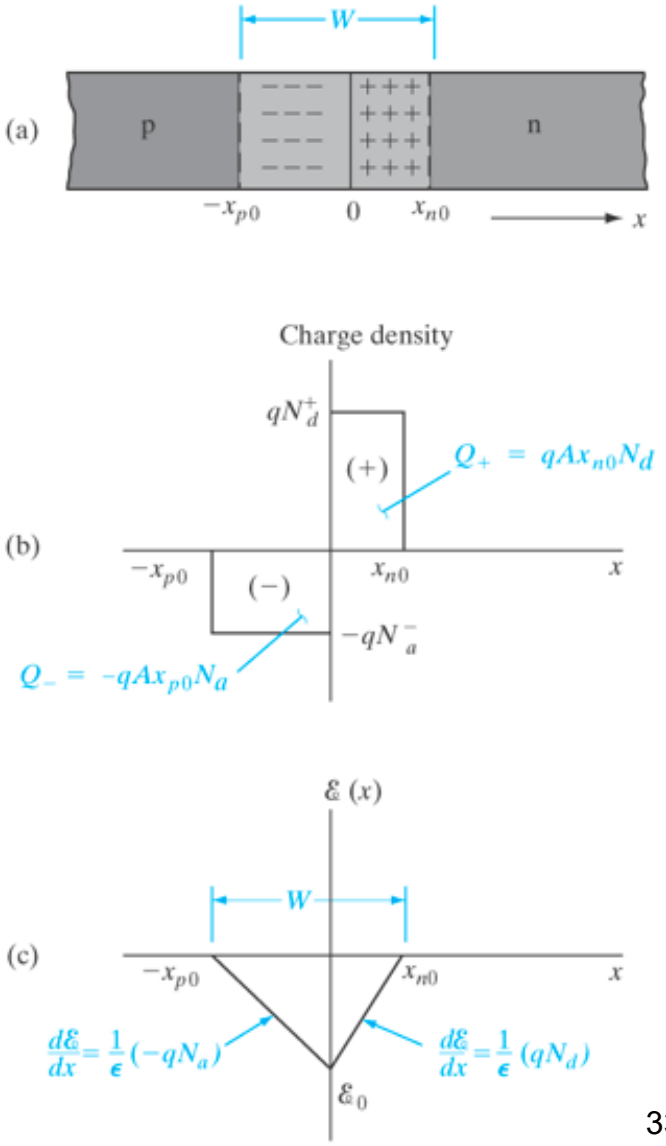
Balance of Charge Requirement: $Q_+ = |Q_-| \Leftrightarrow N_d x_{no} = N_a x_{po}$

$$x_{po} = x_{no}\frac{N_d}{N_a} \text{ and } W = x_{no} + x_{po} = x_{no}\left(1 + \frac{N_a}{N_d}\right),$$

$$\text{so } x_{no} = W\left(\frac{N_a}{N_a + N_d}\right)$$

$$\text{and } V_o = \frac{1}{2}\frac{q}{\varepsilon}N_d x_{no} W = \frac{1}{2}\frac{q}{\varepsilon}N_d\left(W\left(\frac{N_a}{N_a + N_d}\right)\right)W = \frac{1}{2}\frac{q}{\varepsilon}\frac{N_a N_d}{N_a + N_d}W^2$$

$$W = \left[\frac{2\varepsilon V_o}{q}\left(\frac{N_a + N_d}{N_a N_d}\right)\right]^{1/2} = \left[\frac{2\varepsilon V_o}{q}\left(\frac{1}{N_a} + \frac{1}{N_d}\right)\right]^{1/2}$$



Junction Capacitance

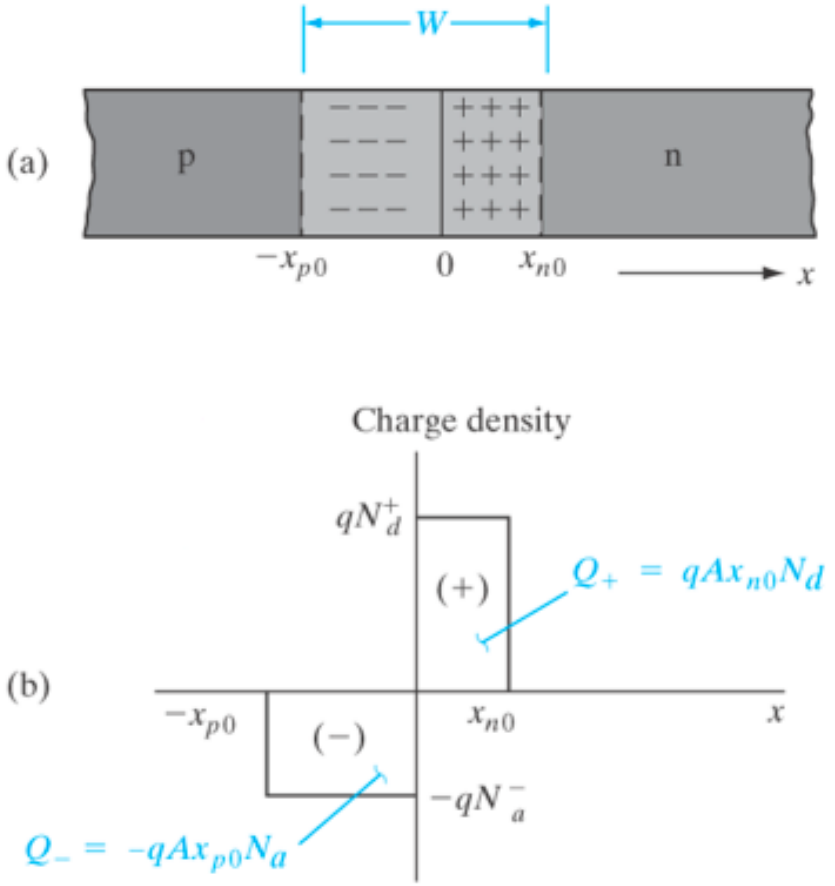
$$C = \left|\frac{dQ}{dV}\right|$$

Under bias: $$W = \left[\frac{2\varepsilon(V_o - V)}{q}\left(\frac{N_a + N_d}{N_a N_d}\right)\right]^{1/2}$$

$$|Q| = qAx_{no}N_d = qAx_{po}N_a$$

$$x_{no} = \frac{N_a}{N_a + N_d}W \text{ and } x_{po} = \frac{N_d}{N_a + N_d}W \text{ so:}$$

$$|Q| = qA\frac{N_d N_a}{N_d + N_a}W = A\left[2q\varepsilon(V_o - V)\left(\frac{N_d N_a}{N_d + N_a}\right)\right]^{1/2}$$



Junction Capacitance [2]

$$|Q| = qA\frac{N_d N_a}{N_d + N_a}W = A\left[2q\varepsilon\left(V_o - V\right)\left(\frac{N_d N_a}{N_d + N_a}\right)\right]^{1/2} \quad \text{define } V' = V_0 - V$$

$$\frac{dV'}{dV} = \frac{d\left(V_0 - V\right)}{dV} = -1 \text{ so } dV' = -dV$$

$$C_j = \left|\frac{dQ}{dV}\right| = \left|\frac{dQ}{-dV'}\right| = \left|\frac{dQ}{-d\left(V_o - V\right)}\right| = \frac{A}{2}\left[\frac{2q\varepsilon}{\left(V_o - V\right)}\left(\frac{N_d N_a}{N_d + N_a}\right)\right]^{1/2}$$

Key Point: A p-n junction has a voltage-variable capacitance

Varactor: device that makes use of this property

$\frac{1}{W}$

Note: $C = \frac{A}{2}\left(\frac{4\varepsilon^2}{4\varepsilon^2}\right)^{1/2}\left[\frac{2q\varepsilon}{\left(V_o - V\right)}\left(\frac{N_d N_a}{N_d + N_a}\right)\right]^{1/2} = \varepsilon A\left[\frac{q}{2\varepsilon\left(V_o - V\right)}\left(\frac{N_d N_a}{N_d + N_a}\right)\right]^{1/2} = \frac{\varepsilon A}{W}$

For a p^+n junction, $N_a \gg N_d$ and $x_{no} \approx W$ while x_{po} is negligible:

$$C_j = \frac{A}{2}\left[\frac{2q\varepsilon}{\left(V_o - V\right)}N_d\right]^{1/2} \quad \leftarrow \quad p^+n$$

Capacitance can be used to measure doping density!

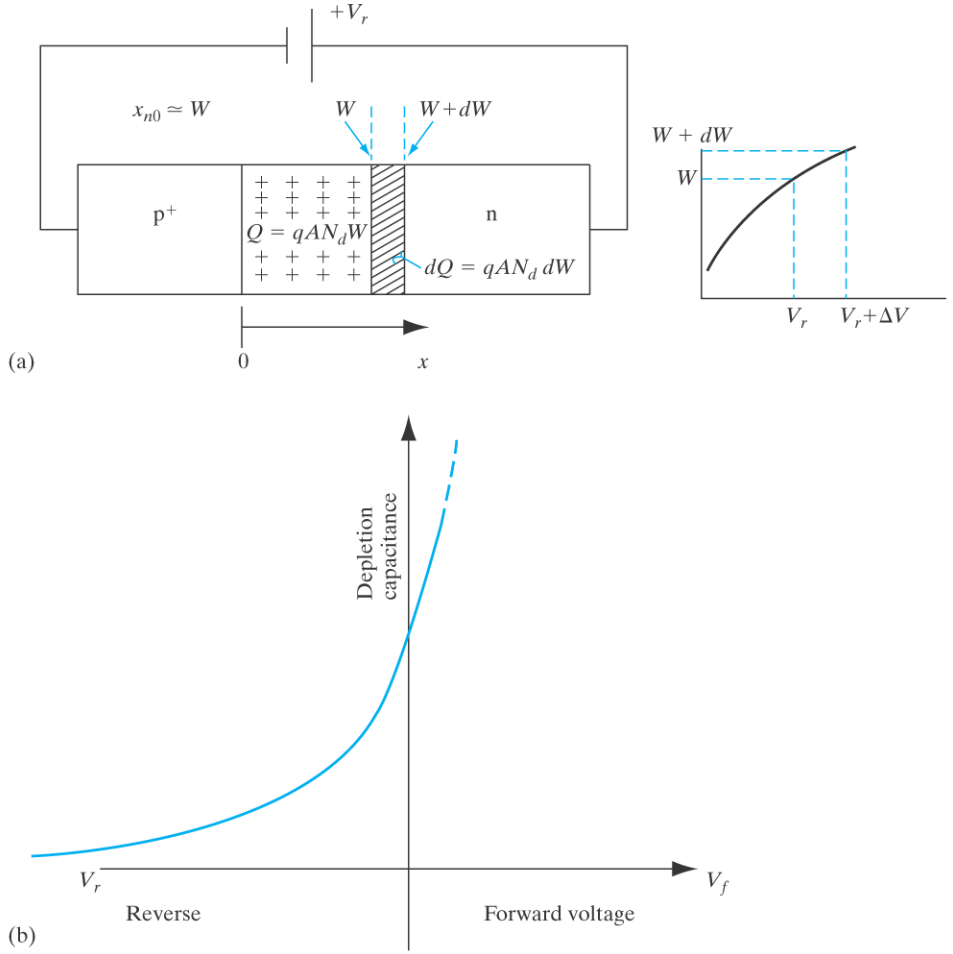


Figure 5.30

Depletion capacitance of a junction: (a) p+-n junction showing variation of depletion edge on n side with reverse bias. Electrically, the structure looks like a parallel plate capacitor whose dielectric is the depletion region, and the plates are the space charge neutral regions; (b) variation of depletion capacitance with reverse bias [Eq. (5-63)]. We neglect x_{p0} in the heavily-doped p+ material.

From *Solid State Electronic Devices,* Sixth Edition, by Ben G. Streetman and Sanjay Kumar Banerjee.
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Example: Depletion Capacitance

For an abrupt Si p-n junction with an area of 10^{-4} cm^{-2} with the properties given in the table, calculate the depletion capacitance at 300 K for a reverse bias of 4V:

p-side	n-side
$N_a=10^{17}$ cm^{-3}	$N_d=10^{15}$ cm^{-3}
$\tau_n=0.1$ μs	$\tau_p=10$ μs
$\mu_p=200$ $cm^2/V\text{-}s$	$\mu_n=1300$ $cm^2/V\text{-}s$
$\mu_n=700$ $cm^2/V\text{-}s$	$\mu_p=450$ $cm^2/V\text{-}s$

$$V_o = \frac{kT}{q}\ln\frac{N_a N_d}{n_i^2} = 0.695V$$

Determine the depletion capacitance at -4V:

$$C_j = \sqrt{\varepsilon}A\left[\frac{q}{2(V_o - V)}\frac{N_d N_a}{N_d + N_a}\right]^{1/2}$$

$$= \sqrt{\left(8.85\times10^{-14}\times11.8\right)}\left(10^{-4}\right)\left[\frac{1.6\times10^{-19}}{2\left(0.695+4\right)}\left(\frac{10^{15}\times10^{17}}{10^{15}+10^{17}}\right)\right]^{1/2}$$

$$= 4.198\times10^{-13}\ \text{F}\ \sim\ 0.42\ \text{pF}$$